

Weekly Diary – Week 9

Internship Organization: Vikaspedia

Duration: 15/03/2026 – 21/03/2026

Objective of the Week

The objective for this week was to strengthen the collaborative editing system by introducing conflict resolution mechanisms, improving overall data consistency, and adding features that enhance user collaboration. The focus was on ensuring that multiple users could work simultaneously on the same document without causing inconsistencies or data loss, while also making the interface more interactive and informative.

Date – 15 March

On this day, advanced concepts related to conflict resolution were studied in detail. Techniques such as Operational Transformation (OT) and Conflict-free Replicated Data Type were explored to understand how modern collaborative systems maintain consistency among multiple users. The study focused on how these methods allow concurrent edits to be merged without conflicts and ensure that all users eventually see the same document state.

Date – 16 March

Building on the theoretical understanding, basic conflict handling logic was implemented in the system. The goal was to manage situations where multiple users attempt to edit the same portion of the document at the same time. Mechanisms were introduced to merge these edits intelligently, ensuring that important data is not overwritten and that all valid changes are preserved effectively.

Date – 17 March

A new feature was added to track users who are actively participating in the editing session. The system was designed to identify and maintain a list of currently connected users. This information was then displayed on the interface, allowing participants to see who else is collaborating on the document in real time.

Date – 18 March

The user interface was further improved by assigning unique identifiers, such as different colors or labels, to each user. This enhancement made it easier to distinguish between contributions made by different collaborators. As a result, users could visually track edits and better understand how the document evolves during collaborative sessions.

Date – 19 March

Work was carried out on maintaining the most up-to-date version of the document on the server side. The backend was enhanced to ensure that every change made by users is accurately stored and reflected in the central document state. Proper synchronization mechanisms were implemented so that all connected clients receive consistent updates.

Date – 20 March

Functionality was added to support new users joining an ongoing editing session. When a new user connects, the system now ensures that they instantly receive the latest version of the document. This eliminates the need for manual refresh and allows new participants to seamlessly become part of the collaboration.

Date – 21 March

Extensive testing was performed involving multiple users editing the document simultaneously. Different scenarios were simulated to evaluate how well the system handles concurrent edits. The testing confirmed that the system maintained consistency, accurate synchronization, and stable performance even under heavy collaborative usage.